崑山科技大學四年制日間部先進應用材料工程系課程科目表

Kun Shan University Department of Advanced Applied Materials Engineering
4-Year College Curriculum (Day Division)

課程名稱(講授時數－實習時數－學分數)
Course Name (Lecture Hours – Practice Hours – Credits)

113.09.25系課程委員會通過Approved by the Department Curriculum Committee
113.11.14院課程委員會通過Approved by the College Curriculum Committee
114.01.21校課程委員會通過Approved by the University Curriculum Committee
114.08.28系課程委員會修正通過Amended and approved by the Departmen Curriculum Committee
114.12.02院課程委員會修正通過Amended and approved by the College Curriculum Committee
115.01.28校課程委員會修正通過Amended and approved by the University Curriculum Committee

114學年度入學適用
Applicable to freshmen enrolled in Fall, 2025

第一學年 (1st Academic Year)		第二學年 (2nd Academic Year)		第三學年 (3rd Academic Year)		第四學年 (4th Academic Year)		
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	暑期 Summer
共同必修科目(合計 28 學分) University-wide Required Courses (28 Credits)								
校訂共同科目(合計 2 學分) General Courses (2 Credits)								
	生活實踐教育(一) Living Practice (I) 0 - 1 - 1	生活實踐教育(二) Living Practice (II) 0 - 1 - 1	服務學習與公民教育 Public Service and Civil Education 0 - 1 - 0					
核心通識科目(合計 16 學分) Common Core Courses (16 Credits)								
運動與生活 (一) Sports and Life (I) 2 - 0 - 1	運動與生活 (二) Sports and Life (II) 2 - 0 - 1	運動專項 (一) Individual Sports Training (I) 2 - 0 - 1	運動專項 (二) Individual Sports Training (II) 2 - 0 - 1	生涯發展與職業倫理(工民商) Career Development and Work Ethics 2 - 0 - 2	外語選讀**(工商) Foreign Languages 2 - 0 - 2			
中文閱讀與表達(一) Chinese Reading and Expression (I) 2 - 0 - 2	中文閱讀與表達(二) Chinese Reading and Expression (II) 2 - 0 - 2							
英文聽力與閱讀(一) English Listening and Reading (I) 2 - 0 - 2	英文聽力與閱讀(二) English Listening and Reading (II) 2 - 0 - 2							
分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)								
分類通識 Liberal Education 2 - 0 - 2	分類通識 Liberal Education 2 - 0 - 2	分類通識 Liberal Education 2 - 0 - 2	分類通識 Liberal Education 2 - 0 - 2	分類通識 Liberal Education 2 - 0 - 2				
8 - 0 - 7	8 - 1 - 8	4 - 1 - 4	4 - 1 - 3	4 - 0 - 4	2 - 0 - 2			

註：Note:
*「外語能力檢定」為本校畢業門檻，依「崑山科技大學學生外語基本能力檢定實施辦法」辦理。
'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'
**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。
'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.
***分類通識包含文史哲藝術領域(A類)、社會科學領域(B類)、自然科學領域(C類)、數位科學領域(E類)、跨域探究領域(F類)，除數位科學領域於一上修習外，其餘領域於畢業前依分類通識規劃分別修習。
General education course are divided into five areas: Humanities (Category A), Social Sciences (Category B), Natural Sciences (Category C), Digital Sciences (Category E), and Cross-disciplinary Studies (Category F). Except for Digital Sciences, which are taken during the first semester of the first year, courses from the other fields must be completed before graduation in accordance with the categorized general education curriculum plan.

院訂基礎必修科目（合計 18 學分）General Compulsory Courses of College (Total：18 Credits)								
微積分(一) Calculus(I) 2 - 0 - 2	微積分(二) Calculus(II) 2 - 0 - 2	工程數學 Engineering Mathematics 2 - 0 - 2						
工程技術概論 Introduction to Engineering Technology 2 - 0 - 2	普通化學(二) General Chemistry(II) 2 - 0 - 2	科技英文 Technical English 2 - 0 - 2						
普通化學(一) General Chemistry(I) 2 - 0 - 2	計算機程式設計 Computer Programming 0 - 2 - 2							
普通物理(一) University Physics(I) 2 - 0 - 2								
8 - 0 - 8	4 - 2 - 6	4 - 0 - 4						

第一學年 (1st Academic Year)		第二學年 (2nd Academic Year)		第三學年 (3rd Academic Year)		第四學年 (4th Academic Year)		
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	暑期 Summer
院訂專業必修科目（合計 18 學分）Professional Modules Compulsory Courses of College (Total：18 Credits)								
普通化學實驗(一) General Chemistry Experiments(I) 0 - 2 - 2	普通化學實驗(二) General Chemistry Experiments(I) 0 - 2 - 2	基礎儀器分析 Instrumental Analysis 1 - 2 - 3	工業安全與衛生 Industrial Safety and Health 2 - 0 - 2	實務專題 Practice Project 0 - 2 - 2	清潔生產 Clean Production 3 - 0 - 3			
	普通物理實驗 Physics Experiments 0 - 2 - 2				專題研究 Special topics 0 - 2 - 2			
0 - 2 - 2	0 - 4 - 4	1 - 2 - 3	2 - 0 - 2	0 - 2 - 2	3 - 2 - 5			
專業必修科目(合計 30 學分) Compulsory Courses (Total: 30 Credits)								
材料科學概論(一) Introduction of Materials Science(I) 3 - 0 - 3		有機化學 Organic Chemistry 3 - 0 - 3	高分子科學 Polymer Science 3 - 0 - 3	材料檢測與分析 Materials Testing and Analysis 1 - 2 - 3	材料加工實驗 Materials Processing and Experiments 1 - 2 - 3	學術倫理* Academic Ethics 1 - 0 - 1	專業證照*** Professional Certificate 0 - 1 - 0	暑期校外實習(一)** Summer out of school internship(I) 0 - * - 2
		半導體材料與元件 Elements and Materials of Semiconductor 3 - 0 - 3	半導體製程與設備 Semiconductor Process and Equipment 3 - 0 - 3	電腦輔助立體製圖與製造 Computer Aided 3D Drawing and Manufacturing 1 - 2 - 3				
			電腦輔助製圖 Computer Aided Drawing 1 - 2 - 3					
3 - 0 - 3		6 - 0 - 6	7 - 2 - 9	2 - 4 - 6	1 - 2 - 3	1 - 0 - 1	0 - 1 - 0	0 - * - 2

註：Note:

1.*「學術倫理」之授課與學分取得依課務組網頁之「學術倫理教育實施辦法」辦理。

The implementation of the “Academic Ethics” course and the acquisition of its credits are carried out in accordance with the Regulations for Academic Ethics Education Implementation on the Academic Affairs Division website.

2.**暑期校外實習課程依材料系之「校外實習必修課程作業要點」辦理。

The summer off-campus internship course is carried out in accordance with the Implementation Guidelines for Off-campus Internship Courses of the Department of Advanced Applied Materials Engineering.

3.***「專業證照」課程依材料系之「專業領域基本能力檢定實施辦法」辦理。

Professional certification courses are carried out in accordance with the Implementation Guidelines for Basic Professional Competency Assessments of the Department of Advanced Applied Materials Engineering.

專業選修科目 (每學期依實際情況開課) Elective Courses (Based on every semester opening courses)								
綠色材料 Green Materials 3 - 0 - 3	材料科學概論(二) Introduction of Materials Science(II) 3 - 0 - 3	基本電學 Basic Electricity 3 - 0 - 3	奈米科技 Nanoscience 3 - 0 - 3	染整實務 Practice of Dyeing and Finishing 1 - 2 - 3	數位印花實務 Practice of Digital Printin 1 - 2 - 3	工廠管理 Industry Management 2 - 0 - 2	學年校外實習(二)** Academic year Internship (II) 0 - * - 9	暑期校外實習(二)** Summer out of school internship(II) 0 - * - 2
	節能材料概論 Introduction to Energy Conservation 3 - 0 - 3	材料力學 Mechanics of Materials 3 - 0 - 3	光電材料 Optoelectronic Materials 3 - 0 - 3	電化學實務 Practice of Electrochemical 2 - 1 - 3	先進複合材料 Advanced Composite Materials 3 - 0 - 3	學年校外實習(一)** Academic year Internship (I) 0 - * - 9	學期校外實習** Semester Internship 0 - * - 9	
	普通物理(二) University Physics(II) 2 - 0 - 2	金屬材料 Metallic Materials 3 - 0 - 3	陶瓷材料 Ceramic Materials 3 - 0 - 3	材料熱力學 Thermodynamics of Materials 3 - 0 - 3	半導體產業概論 Introduction of Semiconductor Industry 3 - 0 - 3		校外職場實習 Internship in Workplace 0 - * - 7	
			織品材料與加工 Fabric materials and processing 1 - 2 - 3	半導體特用化學品 Semiconductor Specialty Chemicals 3 - 0 - 3	高分子材料特性與加工 Polymer Materials Properties and Processing 3 - 0 - 3			
			多媒材與色彩計畫運用 Applications of Multi-media and Color planning 1 - 2 - 3	高分子化性與物性概論 Introduction of Polymer Physics and Chemistry 3 - 0 - 3				
			材料製備實驗 Materials Preparation Experiments 0 - 3 - 3					

註：Note：

**暑期校外實習(一)(二)、學期校外實習、學年校外實習(一)(二)等課程依「崑山科技大學學生校外實習辦法」辦理。

The Summer Off-campus Internship (I)(II), Semester Internship, and Academic Year Internship (I)(II) courses are conducted in accordance with the Regulations for Student Off-campus Internships of Kun Shan University.

最低畢業學分： **128** 學分 Minimum for Total Credits: 128 Credits

第一學年 (1st Academic Year)		第二學年 (2nd Academic Year)		第三學年 (3rd Academic Year)		第四學年 (4th Academic Year)		
第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	暑期 Summer
材料應用數位科技微學程(8學分) 註4 Micro Program in Digital Technology for Materials Applications (8 Credits) Note5								
		通識E General Education E 2 - 0 - 2	電腦輔助製圖 Computer Aided Drawing 1 - 2 - 3	電腦輔助立體製圖與製造 Computer Aided 3D Drawing and Manufacturing 1 - 2 - 3				
綠色材料跨院系學分學程課程(16學分)註5 Interdisciplinary Program in Green Materials (16 Credits) Note6								
材料科學概論(一) Introduction of Materials Science(I) 3 - 0 - 3	節能材料概論 Introduction to Energy Conservation 3 - 0 - 3	綠色能源與產業商機 Green Energy and Opportunity 3 - 0 - 3	環境規劃與管理 Environmental Planning and Management 3 - 0 - 3	材料檢測與分析 Materials Testing and Analysis 1 - 2 - 3	能源應用 Applications of Energy 3 - 0 - 3	基礎儀器分析 Instrumental Analysis 1 - 2 - 3	高分子科學 Polymer Science 3 - 0 - 3	

註：Note：

1.本系最低畢業學分為128學分。

The minimum number of credits required for graduation in this department is 128.

2.暑期校外實習(必修)實習時數至少160小時；學期校外實習 (選修)開設共9學分為連續4.5個月之實習，學年校外實(一)(二) (選修)開設共18學分為連續9個月之實習，相關辦法另訂之。

The Summer Off-campus Internship (compulsory) requires a minimum of 160 hours. The Semester Internship (elective) offers 9 credits and consists of a continuous internship period of 4.5 months. The Academic Year Internship (I)(II) (elective) offers 18 credits and consists of a continuous internship period of 9 months. Relevant regulations are stipulated separately.

3.選修學期校外實習或學年校外實習(一)者得抵免暑期校外實習(一) 2學分且取得該專業選修校外職場實習7學分。

Students who take the Elective Semester Internship or Academic Year Internship (I) may be granted an exemption of 2 credits from Summer Off-campus Internship (I) and earn 7 credits for the Elective Professional Workplace Internship.

4.學生須修畢本系材料應用數位科技微學程課程始得畢業。

Students in this department must complete the Micro Program in Digital Technology for Materials Applications and obtain the corresponding certificate before graduation.

5.學生須修畢本校任一跨院系學程，始得畢業。

Students are required to complete at least one cross-college interdisciplinary program offered by the university in order to graduate.